README

Dataset of reports about MOF-based SERS substrates since 2011 until March 2023.

Juan A. Allegretto[1,*], Jakub Dostalek[1,2,*]

1. *Laboratory for Life Sciences and Technology (LiST), Department of Medicine, Faculty of Medicine and Dentistry, Danube Private University, 3500 Krems, Austria.*
2. *FZU-Institute of Physics, Czech Academy of Sciences, Na Slovance 2, Prague 82021, Czech Republic*

juan.allegretto@dp-uni.ac.at *jakub.dostalek@dp-uni.ac.at*

The dataset utilized in this study was generated from the Web of Science database, encompassing manuscripts published up to March 2023. A literature search was initially conducted using a combination of keywords, including "MOF," "Metal Organic Framework," "SERS," "Surface Enhanced Raman Spectroscopy," and "Surface Enhanced Raman Scattering." This search spanned the "Topic" category, enabling exploration across title, abstract, author keywords, and keyword plus fields.

From the initial pool of 238 documents, review articles and duplicates were systematically excluded, resulting in a refined collection of 182 articles. Subsequently, articles not concurrently addressing MOF and SERS or those utilizing MOF as sacrificial templates were further excluded, resulting in a final subset of 72 articles. From this curated set, relevant parameters were extracted, resulting in 229 entries for the dataset. The extracted information includes the following parameters (in cases where authors did not provide information on a specific parameter, "NR" was recorded, denoting "not reported"):

- Substrate type.
- Employed MOF and configuration.
- Employed metal and configuration.
- Synthesis method or incorporation strategy.
- Chemical modification on the metal.
- Excitation source, wavelength, and power.
- Measurement information, including collection and integration time.
- Analyte of interest and phase.
- Incubation time (and calibration status).
- Performance of interference tests.
- Measurement results in terms of selectivity and reproducibility (RSD).
- Enhancement factor calculation and value.
- Calibration curves and validation experiments.
- Use of internal standards.
- Limit of detection.
- Recovery.